

Design and analysis of experiments

Macartan Humphreys

2026-01-01

Abstract

The course addresses the design and analysis of experiments. It is a hands-on course in which theoretical results are introduced through lecture, demonstrated in practice, and worked on in groups. Topics include randomization schemes, analysis strategies, design evaluation, and workflows.

1 Class resources

- All resources: <https://macartan.github.io/ci>
- This syllabus: https://macartan.github.io/ci/experiments_syllabus_2026.pdf
- Slides: https://macartan.github.io/ci/experiments_2026.html
- [Student survey](#)
- Git repo <https://github.com/macartan/ci>: This repo is for both my causal inference class and this experiments class.

2 Times and locations

Meetings will be held at the WZB

Date	Time	Location	Topic
9 Jan 2026	9 - 13	WZB A 305	Getting started
16 Jan 2026	9 - 13	WZB A 305	Doing experiments
30 Jan 2026	9 - 13	WZB A 305	Causality and causal inquiries
6 Feb 2026	9 - 13	WZB A 305	Answer strategies
20 Feb 2026	9 - 13	WZB A 305	Data strategies and design evaluation
27 Feb 2026	9 - 13	WZB B 001	Topics: downstream experimentation, spillover experiments, patient preference trials, survey experiments

The sessions will be structured roughly as follows.

- 9:00 - 11:00: lecture or discussions of readings
- 11:00 - 11:15: break
- 11:15 - 13:00: discussions, team presentations and in-class exercises

3 Expectations

3.1 Required

- Do the readings and take part in discussion (25%)
- Take part in field experiment I (25%)
- Design an experiment as a candidate for survey experiment II (25%)
- Short in class exam (25%)

3.2 Optional

- Prepare a research design or short paper, perhaps building on existing work. Typically this contains:
 - a problem statement
 - a description of a method to address the problem
 - analytic or simulation based results describing properties of the solution
 - a discussion of implications for practice.
- A passing paper will illustrate subtle features of a method; a good paper will identify unknown properties of a method; an excellent paper will develop a new method.

4 Readings

The course combines lectures, discussion, and activities.

All readings are linked from this document.

I will draw material especially from Blair, Coppock, and Humphreys (2023): [Research Design in the Social Sciences](#)

Recommended but not open access: [Gerber and Green \(2012\)](#) with supplementary material available here: <https://isps.yale.edu/FEDAI>

Readings for each session are given below.

4.1 Intro

Aims:

- Introduce the class
- Practice running experiments by hand
- Introduce and discuss field experiment I
- To prepare for next session: proposed adaptation of field experiment 1

Course outline, tools

4.2 Experimentation

Aims:

- Review lifecycle of an experiment (discussion)
- Discuss ethical challenges to experimentation
- Develop group field experiment 1

Required reading:

- [The Hijab penalty](#) and the [Appendix](#)
- [Johnson: Protecting the Community: Lessons from the Montana Flyer Project](#)
- [DD, Ch 2](#) introduces ideas at a high level
- [Design lifecycle](#) Blair, Coppock, and Humphreys (2023) p 319 - 358
- [Humphreys \(2015\)](#) on ethics in experiments

Optional reading:

- [Alvarez, Key, and Núñez \(2018\)](#)
- [Humphreys, De la Sierra, and Van der Windt \(2013\)](#) on fishing and registration

4.3 Causality

Aims:

- Understand the potential outcomes framework
- Understand identification challenges and the role of randomization
- Understand varieties of estimands

Required:

- [Holland](#) is a beautiful classic reading on the fundamental problem of causal inference
- [DD, Ch 7](#)

Optional:

- [II Ch 2](#), goes over both potential outcomes and DAGs
- [Keele \(2015\)](#) gives a good high level overview of many key ideas in this course
- [Imai, King, and Stuart \(2008\)](#)
- [Hernán and Robins \(2024\)](#) (Section 3.1, 7.2, 8.4, 10.1)
- [Hernán and Robins \(2024\)](#) Ch 6
- [II Ch 4](#)
- [Lundberg, Johnson, and Stewart \(2021\)](#)

4.4 Answer strategies

Aims:

- Understand alternative strategies to estimating treatment effects
- Understand design based strategies for estimating standard errors and p -values
- Understand confidence intervals
- Understand the role of covariates in experimental design

Required:

- [Freedman \(2008\)](#) helps make connections between design based inference and regression
- [Lin \(2012\)](#) relieves some worries you might have after reading Freedman (2008)

Optional:

- [Bind and Rubin \(2020\)](#) on RI
- [Chang, Middleton, and Aronow \(2024\)](#) follows up on Lin (2012).

4.5 Experimental Design and Evaluation

Aims:

- Understand alternative assignment strategies
- Understand statistical power
- Understand how to assess alternative diagnostics

Required:

- [DD, Ch 7](#)
- [DD, Ch 8](#) on data strategies
- [DD, Ch 9](#) on answer strategies
- [DD, Ch 13](#) gives more practical guidance on getting started

Optional:

4.6 Topics

Aims

- Topics TBD

Required reading (TBD):

4.6.1 downstream experimentation

4.6.2 spillover experiments

- [Aronow and Samii \(2017\)](#) on spillovers

4.6.3 patient preference trials

4.6.4 survey experiments

4.6.5 mediation experiments

- [Imai, Keele, and Tingley \(2010\)](#) on mediation

5 Prerequisites and tools

You should already have background in statistics up to the point of feeling comfortable with regression.

In addition you should know some R. Really, the more you can invest on getting on top of R before the class the better.

Before the first class please make sure your R is up-to-date and that you are working in R studio. Then make sure you have the following packages installed.

```
pacman::p_load(
  rstan,
  dagitty,
  DeclareDesign,
  CausalQueries,
  tidyverse,
  knitr
)
```

Resources:

- Resources for learning R: <http://www.r-bloggers.com/how-to-learn-r-2/>
- *Hadley Wickham's [R for Data Science](#)
- [Grant McDermott on data science](#)

5.1 Writing with qmd

Please plan to do drafting in Quarto. This is a simple markup language that lets you integrate writing and coding. This document is written in quarto and the slides will be also.

The key thing is that you can insert code chunks like this.

```
# Define a random number
x <- rnorm(1)
```

Code like this is run as the document compiles, and results can be accessed as needed, like this: we just sampled the random number $x = 1.2633152$.

- See [Wickham, Çetinkaya-Rundel, and Golemund \(2023\)](#) on [Workflow](#) and [Quarto](#)

I recommend using Rstudio as an editor. More information here: <https://quarto.org/docs/get-started/hello/rstudio.html>

5.2 AI

AI can be a great aid to help understand this material and also to develop code. However it should only be used as an aid: it makes many many mistakes; you remain responsible for your code and more importantly for your comprehension.

References

- Alvarez, R Michael, Ellen M Key, and Lucas Núñez. 2018. “Research Replication: Practical Considerations.” *PS: Political Science & Politics* 51 (2): 422–26.
- Aronow, P M, and Cyrus Samii. 2017. “Estimating Average Causal Effects Under General Interference, with Application to a Social Network Experiment.” *Annals of Applied Statistics* 11 (4): 1912–47.
- Bind, M-AC, and DB Rubin. 2020. “When Possible, Report a Fisher-Exact p Value and Display Its Underlying Null Randomization Distribution.” *Proceedings of the National Academy of Sciences* 117 (32): 19151–58.
- Blair, Graeme, Alexander Coppock, and Macartan Humphreys. 2023. *Research Design in the Social Sciences: Declaration, Diagnosis, and Redesign*. Princeton University Press.
- Chang, Haoge, Joel A Middleton, and PM Aronow. 2024. “Exact Bias Correction for Linear Adjustment of Randomized Controlled Trials.” *Econometrica* 92 (5): 1503–19.
- Freedman, David A. 2008. “On Regression Adjustments to Experimental Data.” *Advances in Applied Mathematics* 40 (2): 180–93.

- Gerber, Alan S, and Donald P Green. 2012. *Field Experiments: Design, Analysis, and Interpretation*. Norton.
- Hernán, Miguel A, and James M Robins. 2024. *What If?* Boca Raton: Chapman & Hall/CRC.
- Humphreys, Macartan. 2015. “Reflections on the Ethics of Social Experimentation.” *Journal of Globalization and Development* 6 (1): 87–112.
- Humphreys, Macartan, Raul Sanchez De la Sierra, and Peter Van der Windt. 2013. “Fishing, Commitment, and Communication: A Proposal for Comprehensive Nonbinding Research Registration.” *Political Analysis* 21 (1): 1–20.
- Imai, Kosuke, Luke Keele, and Dustin Tingley. 2010. “A General Approach to Causal Mediation Analysis.” *Psychological Methods* 15 (4): 309–34.
- Imai, Kosuke, Gary King, and Elizabeth A Stuart. 2008. “Misunderstandings Between Experimentalists and Observationalists about Causal Inference.” *Journal of the Royal Statistical Society Series A: Statistics in Society* 171 (2): 481–502.
- Keele, Luke. 2015. “The Statistics of Causal Inference: A View from Political Methodology.” *Political Analysis* 23 (3): 313–35.
- Lin, Winston. 2012. “Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman’s Critique.” *arXiv Preprint arXiv:1208.2301*.
- Lundberg, Ian, Rebecca Johnson, and Brandon M Stewart. 2021. “What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory.” *American Sociological Review* 86 (3): 532–65.
- Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science*. O’Reilly Media, Inc.